RESTRICTION OF AN INJECTIVE MODULE ON A RINGED SPACE TO A SMALLER OPEN IS SURJECTIVE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed. The counted statement opens with the author's own slogan 'Injectives are flasque'.

Editorial note. The chapter's retained preamble does not carry the project's no-op definitions of the display environments, so a second verbatim preamble span supplies the project's own no-op definitions of reference, slogan and history, taken from the snapshot at sources/stacks-varieties/retained-preamble.tex lines 75-77 (themselves the project's wrapper definitions). Nothing was invented and no macro was redefined.

Editorial note. All four referenced results live in earlier chapters, so their locators point at the verbatim lines of the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt) that carry the canonical tags.

Allowed standard prerequisites (editorial).

- Extension by zero of the structure sheaf along an open immersion and its left adjointness to restriction. Sheaves, Section sheaves-section-open-immersions (tag 009Z) and Sheaves, Lemma sheaves-lemma-j-shriek-modules (tag 00A7); invoked at proof lines 728-740. Supply that j_! O_U represents the functor F -> F(U) and that restriction is its right adjoint; the Yoneda comparison is retained in the counted span.
- Yoneda's lemma. Categories, Lemma labelled categories-lemma-yoneda (canonical Stacks tag 001P); invoked at proof line 750. Supplies that the canonical map j'_! O_{U'} -> j_! O_U corresponds to the restriction map on U-sections.
- Characterisation of injective objects by surjectivity of Hom out of a monomorphism. Homology, Lemma labelled homology-lemma-characterize-injectives (canonical Stacks tag 0136); invoked at proof lines 761-762. Supplies the final step: injectivity of I turns the monomorphism j'_! O_{U'} -> j_! O_U into a surjection on Hom groups, which is the asserted surjectivity of the restriction map.

Source reference. Original section 009Z: sources/stacks-chow/tags.txt, source lines 371–371. Cross-chapter pointer invoked once at proof lines 729-730 for the extension by zero of O_U; the adjointness computation and the Yoneda step are retained in the counted span.

Source reference. Original lemma 00A7: sources/stacks-chow/tags.txt, source lines 379–379. Cross-chapter standard prerequisite invoked once at proof line 740 for the displayed identification Hom(j_! O_U, F) = F(U); the comparison of the two representing sheaves is retained in the counted span.

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Source reference. Original lemma 001P: sources/stacks-chow/tags.txt, source lines 70–70. Cross-chapter standard prerequisite invoked once at proof line 750 to identify the canonical map of representing sheaves with the restriction map on sections.

Source reference. Original lemma 0136: sources/stacks-chow/tags.txt, source lines 1424–1424. Cross-chapter standard prerequisite invoked once at proof lines 761-762 for the criterion turning the monomorphism into a surjection on Hom groups; the final identification with the restriction map is retained in the counted span.

Lemma 0.1. *Injectives are flasque. Let X be a ringed space. Let $U' \subset U \subset X$ be open subspaces. For any injective $\mathcal{O}_X$-module $\mathcal{I}$ the restriction mapping $\mathcal{I}(U) \to \mathcal{I}(U')$ is surjective.*

Proof. Let $j : U \to X$ and $j' : U' \to X$ be the open immersions. Recall that $j_!\mathcal{O}_U$ is the extension by zero of $\mathcal{O}_U = \mathcal{O}_X|_U$, see Sheaves, Section 009Z. Since $j_!$ is a left adjoint to restriction we see that for any sheaf $\mathcal{F}$ of $\mathcal{O}_X$-modules

$$\mathrm{Hom}_{\mathcal{O}_X}(j_!\mathcal{O}_U, \mathcal{F}) = \mathrm{Hom}_{\mathcal{O}_U}(\mathcal{O}_U, \mathcal{F}|_U) = \mathcal{F}(U)$$

see Sheaves, Lemma 00A7. Similarly, the sheaf $j'_!\mathcal{O}_{U'}$ represents the functor $\mathcal{F} \mapsto \mathcal{F}(U')$. Moreover there is an obvious canonical map of $\mathcal{O}_X$-modules

$$j'_!\mathcal{O}_{U'} \longrightarrow j_!\mathcal{O}_U$$

which corresponds to the restriction mapping $\mathcal{F}(U) \to \mathcal{F}(U')$ via Yoneda's lemma (Categories, Lemma 001P). By the description of the stalks of the sheaves $j'_!\mathcal{O}_{U'}$, $j_!\mathcal{O}_U$ we see that the displayed map above is injective (see lemma cited above). Hence if $\mathcal{I}$ is an injective $\mathcal{O}_X$-module, then the map

$$\mathrm{Hom}_{\mathcal{O}_X}(j_!\mathcal{O}_U, \mathcal{I}) \longrightarrow \mathrm{Hom}_{\mathcal{O}_X}(j'_!\mathcal{O}_{U'}, \mathcal{I})$$

is surjective, see Homology, Lemma 0136. Putting everything together we obtain the lemma. □